

Crypto Portfolio Tracker — Setup Guide

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Welcome! This guide walks you through everything you need to get your portfolio syncing automatically. Total time: **~10 minutes** for the Notion side, **~5 minutes** for the sync script.

You only need to do this **once**. After setup, new wallets you add to Notion will sync automatically.

What you'll set up

1.  Duplicate the Notion template into your workspace
 2.  Get three free API keys (Moralis, Helius, Notion integration)
 3.  Run the sync script on any computer or VPS
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Step 1 — Duplicate the template

1. Open the Notion template link from your Gumroad receipt
 2. Click **Duplicate** in the top-right corner
 3. Choose your workspace — the template (11 databases + dashboard) will be copied in
 4. Rename the top page to anything you like (e.g., "My Portfolio")
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Step 2 — Get your free API keys

You need three keys. All are free. Skip Helius if you don't hold any Solana tokens; skip Moralis if you only hold Bitcoin + Solana.

2a. Moralis (EVM chains — Ethereum, Arbitrum, Base, etc.)

1. Go to deep-index.moralis.io and sign up (free)
2. Create a new project
3. Navigate to **Settings** → **API Keys**
4. Copy your **Web3 API Key**
5. Free tier: 40,000 compute units/day — enough for a personal portfolio of ~5 wallets scanned every 15 minutes

2b. Helius (Solana)

1. Go to helius.dev and sign up (free)
2. Create a new project
3. Copy your **API Key** from the dashboard
4. Free tier: 1,000,000 credits/month — plenty for personal use

2c. Notion Integration

1. Go to notion.so/my-integrations
2. Click **+ New integration**
3. Name it anything (e.g., “Portfolio Sync”)
4. Associated workspace: pick the one with your duplicated template
5. Capabilities: check **Read content**, **Update content**, **Insert content**
6. Click **Submit**
7. Copy the **Internal Integration Secret** (starts with ntn_ or secret_)

2d. Connect the integration to your template

1. Open your duplicated Notion template (the top-level page)
 2. Click the ... menu in the top-right
 3. Scroll down to **Connections** → **Connect to** → pick your new integration
 4. A popup confirms — hit **Confirm**
 5. This grants the integration access to the top page and all its subpages/databases
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Step 3 — Run the sync script

You have three options. Pick whichever fits your comfort level.

Option A — Run locally (easiest, free, no VPS)

Requirements: a computer that's on when you want portfolio updates (desktop, laptop, Mac Mini, old Raspberry Pi).

1. Install Python 3.10+ if you don't already have it
2. Unzip the automation-bonus.zip from your Gumroad receipt — you'll see scanner.py, sync_daemon.py, and db_ids.json
3. Create a file named .env in the same folder, with these three lines:

```
MORALIS_API_KEY=your_moralis_key_here
HELIUS_API_KEY=your_helius_key_here
NOTION_TOKEN=your_notion_integration_secret_here
```

4. You also need db_ids.json — **this is unique to your workspace**. Run build_databases.py once to generate it:

```
pip install requests
python3 build_databases.py https://www.notion.so/your-template-
page-url
```

Wait — actually you don't need this if you're using the duplicated template. Skip to 4b.

4b. Run get_db_ids.py (included) — it walks your workspace and writes the correct db_ids.json: `bash python3 get_db_ids.py https://www.notion.so/your-template-page-url`

1. Now start the sync daemon:

```
python3 sync_daemon.py
```

You should see sync daemon starting — poll=60s stale=15m in the terminal.

2. Go add a wallet in the Notion **Wallets** database: fill in Label, Address, Chain, and tick **Scan Enabled**. Within 60 seconds, the **Holdings** database will populate.

Option B — Run on a free VPS (always-on)

Best for: 24/7 sync without leaving a laptop running.

Free options: [Oracle Cloud Free Tier](#) (AArch64, 4GB RAM, free forever), [Fly.io](#) (3 small VMs free), [Google Cloud Free Tier](#) (e2-micro, free forever).

1. Spin up a small VM (any Linux distro)
2. SSH in, install Python 3 + git

3. Copy the automation-bonus files to the VM
4. Follow Option A steps 3–5
5. To run in the background:

```
# quick and dirty:
nohup python3 sync_daemon.py > sync.log 2>&1 &

# or as a systemd service (recommended):
sudo cp crypto-sync.service /etc/systemd/system/
sudo systemctl enable --now crypto-sync
sudo systemctl status crypto-sync
```

The `crypto-sync.service` file is included in the automation bonus.

Option C — Run as an n8n workflow

Best for: users already running n8n for other automations.

1. Import `wallet-scanner.json` (included in the automation bonus) into your n8n instance
 2. Open the workflow → click each HTTP node → paste your Moralis / Helius / Notion credentials
 3. Activate the workflow
 4. Same behavior as the Python daemon: polls your Wallets DB every 5 minutes
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Step 4 — Verify it's working

1. Open your Notion **Wallets** database
 2. Add a row with any wallet address you want to test. Example test wallets:
 - EVM: `0xcB1C1FdE09f811B294172696404e88E658659905`
 - Solana: `5Q544fKrFoe6tsEbD7S8EmxGTJYAKtTVhAW5Q5pge4j1`
 - Bitcoin:
`bc1qqdjv0av3q56jvd82tkdjpy7gdp9ut8tlqmgrpmv24sq90ecnvqqjwv`
`w97`
 3. Tick **Scan Enabled**
 4. Watch the **Holdings** database — fresh rows should appear within ~60 seconds
 5. The wallet's **Last Scanned** and **Scan Status** fields update automatically
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Common issues

Problem	Fix
"NOTION_TOKEN missing"	You forgot the .env file or have a typo. Check the file is in the same folder as sync_daemon.py.
"Could not find object" from Notion	You didn't share the page with your integration. Go back to Step 2d.
"Request is unauthorized" from Moralis	API key is wrong or your account is over the free tier. Double-check the key.
Holdings DB is empty	Scan Enabled is unticked, OR the address is wrong, OR the script isn't running. Check the sync log.
Sync is slow	Free tier Moralis limits scans to ~6 EVM chains. A 5-wallet portfolio takes ~15-30 seconds per full cycle.
I removed a wallet but Holdings still show it	The daemon archives orphans on the next cycle (within 60s). If not, run <code>python3 cleanup_orphans.py --apply</code> .

How the formulas work

All P&L math is done by Notion formula fields — you never need to calculate manually.

- **Holdings → P&L USD** = Current Value USD - (Amount × Avg Cost USD)
- **Holdings → P&L %** = P&L USD ÷ cost basis
- **DeFi Positions → Realized Yield USD** = Current Value + Rewards Earned - Initial Deposit
- **Watchlist → Entry Signal** =  BUY when Current Price ≤ Target Entry, otherwise  wait
- **Watchlist → Distance to Entry %** = percentage gap between current and target

To get accurate P&L, make sure you fill in **Avg Cost USD** on each Holdings row. The sync script doesn't know your cost basis — it just writes the current amount and price. You set the cost basis either: - Manually when you buy (quickest) - Automatically by logging every Buy/Sell in the Transactions database with the Tax Lot Method of your choice

Privacy & security

- **Wallet addresses are read-only.** The sync script only queries public blockchain balance APIs — it can never move or sign anything.
 - **API keys stay on your machine** — nothing goes to our server. We have zero visibility into your holdings.
 - **Notion integration is scoped to the template.** Unshare the page at any time to revoke access.
 - **The sync script is open source** — you can read every line in `sync_daemon.py` before running it.
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Getting help

- Read this guide again carefully — 90% of issues are typos in the `.env` file or a missed Notion integration share
- Check the sync log (`sync_daemon.log` in the script folder)
- Reply to your Gumroad receipt email — you'll get an answer within 24 hours
- Feature requests and bug reports welcome

Happy tracking! 📊